

Bernardo Lin

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EDUCATION

University of Washington

Seattle, WA

B.S. in Electrical and Computer Engineering, Dean's List

Sep 2023 - Expected Dec 2026

- Relevant Coursework: Computer Architecture Series, VLSI Series, Data Structures and Algorithms, Embedded Systems, Digital Circuits and Systems Series, Circuit Analysis, Programming for Signal Processing

EXPERIENCE

Hardware Research Intern

Jul 2025 – Sep 2025

Academia Sinica

Taipei, TW

- Debugged **six** open-source **PIM simulators**, resolving legacy code issues and extending **PIMSimulator** with **inter-DPU communication** to model a **2x2 mesh topology** for **graph pattern mining(GPM)** workloads
- Conducted a **literature review** of **25 papers** on **GPM in PIM architectures** and designed a **generalized high-performance streaming GPM workflow**, **validated** by faculty review and **adopted** for ongoing research
- Executed **benchmark testing** on simulators, improving **computational efficiency by 20%** through **algorithm optimizations**, and developed **custom benchmark** for **verification and performance evaluation**

Undergraduate Research Assistant

Nov 2024 – Present

PN Computer Engineering Lab

Seattle, WA

- Worked on reproducing the **Riptide CGRA** architecture in **SystemVerilog** from scratch, as presented in the **IEEE MICRO 2022 paper**, completing different types of **Processing Elements(PEs)** and developed parts of **ucore** while independently resolving unaddressed **low-level design decisions**
- Prototyped the **HyCube** architecture by using the Domain Optimized Reconfigurable Architecture(DORA) interface and enriched **primitive APIs** within DORA's PE Builder class

PROJECTS

FPGA NPU Accelerator | *SystemVerilog, AXI, Systolic Array, Testbench Design, Wave Simulation*

Jun 2026

- **Architected** and **implemented** an **FPGA-based NPU** in **SystemVerilog RTL** targeting **INT8 inference of YOLOv8n**, using a parameterized **INT8 weight-stationary systolic array** accelerator for efficient matrix multiplication in neural networks.
- **Designed** control and execution logic including an **instruction sequencer**, **requant pipeline**, and a **VPU (Vector Processing Unit)** for post-processing operations such as activation functions, bias addition, pooling/reduction, and other neural network inference stages.
- Developed **high-bandwidth** data movement and memory infrastructure using **AXI/AXI-DMA**, **double-buffered BRAM/SRAM**, and **tiled buffer scheduling** to stream weights and activations efficiently during inference execution.

Conway's Game of Life ASIC Accelerator | *SystemVerilog, RTL Design, ASIC Flow, Verification*

Jun 2026

- **Designed** and **implemented** **SystemVerilog RTL** for a parameterized **32x32 Conway's Game of Life ASIC accelerator**, including compute-cell **datapath logic**, **synchronous state updates**, and **FSM-based control** for frame execution
- **Ran the full ASIC design flow** through **RTL simulation**, **synthesis**, **post-synthesis simulation**, **place-and-route**, **post-PAR simulation**, **timing/PPA analysis**, and **DRC/LVS signoff validation**.

Register File Design | *Cadence Virtuoso, Custom CMOS Layout, Calibre DRC/LVS, Digital Circuits*

Nov 2025

- **Designed and verified** a **16-bit, 13-entry CMOS register file** with dual asynchronous read ports and synchronous write port, achieving **1 GHz minimum timing** using fully static CMOS bit-cells, buffers, decoders, and mux logic.
- **Implemented** complete **schematic-to-layout top-level integration**, achieving **DRC/LVS-clean hierarchical layout** within the **200µm × 70 µm die constraint** and characterizing **timing, energy and post-layout PPA** with **HSPICE**

TECHNICAL SKILLS

Software: Quartus, Modelsim, Cadence Virtuoso, Synopsys Verdi, HSPICE, Linux, SystemVerilog, Tcl, C++, Python, Git

Hardware: FPGAs(Altera DE1-SoC, Xilinx KR260), PCBs, Microcontrollers(nrf52 series, RasPi4, stm32, Teensy 4.1)

Electronics: Oscilloscope, Multimeter, Waveform Generator, DC power supplies, Soldering